A Reproducible Colonoscopy Classification Benchmark: Mask-Aware SE-ResNet with CNN-JEPA

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Abstract

Computer-aided analysis of colonoscopy frames can support polyp detection and quality assessment, but labeled data remain scarce and evaluation protocols are fragmented across datasets and architectures. We contribute a **reproducible colonoscopy classification benchmark**: mask-aware SE-ResNet-50 with convolutional JEPA (CNN-JEPA) pretraining, prepare scripts, and a unified evaluation protocol (macro-F1, per-class recall). We report two completed tasks on public Simula data: (T3) three-class screening (landmark / normal mucosa / polyp; C0–C10) and (T1) HyperKvasir 23-class fine-grained taxonomy (B2–B6). We compare ResNet-50, SE-ResNet-50, dual-stream color/texture fusion, and CNN-JEPA fine-tuning under the same training defaults. **We do not claim that domain CNN-JEPA beats ImageNet transfer on these public splits.** On T3 (image-level split, single center), ImageNet transfer reaches macro-F1 0.989 and polyp recall 0.971 (C1), versus macro-F1 0.910 and polyp recall 0.829 for CNN-JEPA fine-tuning (C4). On T1, macro-F1 $\approx$0.44–0.59 reflects severe class imbalance; dual-stream ImageNet (B6) matches ResNet ImageNet (B2) within measurement noise (0.586 vs 0.585). CNN-JEPA shows a label-efficiency *trend* on T3 but does not surpass ImageNet at any tested label fraction. We release code, JSON results, and auto-generated LaTeX tables; grouped splits and multi-seed reporting are planned extensions. **Keywords:** colonoscopy, classification benchmark, self-supervised learning, JEPA, SE-ResNet, HyperKvasir.

1 Introduction

Colonoscopy is the gold standard for colorectal cancer screening. Computer-aided detection (CAD) systems analyze individual frames to flag suspected polyps and to confirm anatomical landmarks required for quality metrics. Deep convolutional networks are attractive for near real-time deployment, yet they require large labeled datasets, and published results are hard to compare across architectures, pretraining regimes, and task definitions.

Public benchmarks mix binary polyp detection, collapsed three-class screening, and fine-grained multi-class taxonomies without a shared protocol or codebase. Self-supervised pretraining may reduce annotation cost, but its benefit over ImageNet transfer for endoscopy remains an empirical question—not a settled claim.

Joint-Embedding Predictive Architectures (JEPA) [6; 2] learn by predicting latent representations of masked views. We adapt a *convolutional* JEPA to colonoscopy frames with mask-aware squeeze-and-excitation (SE) blocks compatible with spatial masking during pretraining.

> **In brief.** We provide an open, reproducible **benchmark and baseline implementation**—not a claim that domain JEPA beats ImageNet on public Simula data. Our primary contribution is the protocol, code, and honest reporting on completed tasks T3 (screening) and T1 (HyperKvasir-23).

Contributions. (1) **Method:** CNN-JEPA on SE-ResNet-50 with mask-aware channel attention for masked pretraining. (2) **Benchmark:** reproducible C0–C10 (T3) and B2–B6

(T1) runs with early stopping, JSON metrics, and auto-generated tables. (3) **Empirical study:** ImageNet transfer outperforms domain CNN-JEPA on the current public splits; we document label-efficiency trends and ablations without overstating gains.

2 Related Work

Endoscopy classification. Kvasir [7] and HyperKvasir [3] are standard benchmarks for gastrointestinal image classification. Prior work in our repository [1] compared ResNet-50, SE-ResNet-50, and dual-stream color–texture fusion on HyperKvasir (23 classes), reaching test macro-F1 $\approx$0.58–0.59 (B2–B6). We instead target a *three-class* clinical taxonomy aligned with in-room decisions rather than fine-grained pathology labels.

Channel attention. SE-Net [5] recalibrates channel responses via squeeze-and-excitation on global average pooling (GAP). Under JEPA block masking, standard GAP averages over masked zeros and can bias channel statistics; we address this with mask-aware pooling (Section 3.2).

JEPA and self-supervision. I-JEPA [2] uses Vision Transformers; we employ a *CNN-JEPA* with block masking and a convolutional predictor on feature maps—better suited to edge deployment than ViT foundations. We do not claim to invent convolutional latent-prediction pretraining; our novelty is the **mask-aware SE integration** and **reproducible benchmark protocol** for colonoscopy classification. Related latent-prediction methods include BYOL [4]. Medical self-supervised literature covers other modalities; we position colonoscopy CNN-JEPA as an empirical baseline within a unified benchmark.

Polyp detection. Classical CAD systems [8] emphasize polyp *sensitivity*. We report polyp recall alongside macro-F1. Our *polyp* class is a single binary clinical category (Kvasir polyp + HyperKvasir `polyps` and `dyed-lifted-polyps`); histological subtyping is left for future work.

3 Methods

3.1 CNN-JEPA pretraining

Given an unlabeled frame $x \in \mathbb{R}^{3\times H\times W}$, a binary block mask $m \in \{0,1\}^{1\times H\times W}$ defines visible context $x \odot m$. A *context encoder* f_θ (SE-ResNet-50) produces a feature map $h = f_\theta(x \odot m, m)$. A lightweight convolutional *predictor* g_ϕ maps h to a latent grid $\hat{z} = g_\phi(h)$. A *target encoder* $f_{\bar{\theta}}$ (EMA copy of f_θ, momentum $\tau = 0.996$) encodes the full image: $z = f_{\bar{\theta}}(x)$. The training loss is normalized mean squared error:

$$\mathcal{L}_{\text{JEPA}} = \|\text{norm}(\hat{z}) - \text{norm}(z)\|_2^2 . \tag{1}$$

Block masks expose 25–45% of pixels (multi-scale rectangles). Pretraining uses AdamW, cosine schedule, early stopping on validation latent loss (patience 10, max 80 epochs).

3.2 Mask-aware squeeze-and-excitation

Standard SE blocks apply GAP over all spatial locations. When inputs are masked, zero-padded regions skew channel statistics. Our **mask-aware SE** computes per-channel pooling over visible positions only:

$$\text{GAP}_c(x, m) = \frac{\sum_{i,j} x_{c,i,j}\, m_{i,j}}{\sum_{i,j} m_{i,j} + \epsilon}. \tag{2}$$

Channel attention weights are then applied as in SE-Net [5]. Ablation C3 uses standard SE under the same JEPA pipeline; C4 enables mask-aware SE.

3.3 Data and class taxonomy

T3 (screening). We merge Kvasir v1 and HyperKvasir labeled images into three classes:

- **Landmark:** cecum, pylorus, z-line, ileum, retroflexions.
- **Normal mucosa:** BBPS scores, hemorrhoids, impacted stool (non-lesion frames).
- **Polyp:** Kvasir `polyp`, HyperKvasir `polyps` and `dyed-lifted-polyps`.

Total: 9565 images, center `simula`. Splits use **group-aware** stratification (procedure/video group per image; `prepare-colonoscopy-3class` reads HyperKvasir `image-labels.csv`). Current public stills are one frame per group; multi-frame video groups apply when unlabeled frame pools are used.

T1 (HyperKvasir-23). Official Simula 23-class folders via `prepare-hyperkvasir`; 10,662 images, stratified 70/15/15.

Planned extensions (not in main tables): T2 pathology-11 subset; JEPA fine-tunes on HK-23.

3.4 Experimental protocol

T3 (C0–C10). Screening three-class runs with label fractions C7–C9.

T1 (B2–B6). ResNet, SE-ResNet, dual-stream on HyperKvasir-23 (supervised only in this manuscript).

3.5 Metrics

Primary: **macro-F1** on validation/test. Clinical secondary: **polyp recall** = TP/(TP + FN) for the polyp class (sensitivity among true polyps).

3.6 Implementation

PyTorch on Apple MPS; open repository `endoscopy-resnet`. Campaign orchestration: `run_full_jepa_benchm` early-stop logs per run.

4 Experiments

Completed tasks.

- **T3 (screening):** three-class colonoscopy (C0–C10), single public center `simula`.
- **T1 (fine-grained):** HyperKvasir 23-class supervised baselines (B2–B6).

Extended tasks (pathology-11, JEPA on HK-23, ViT linear probe) are defined in the repository roadmap but *not* reported here until grouped splits and multi-seed runs complete.

Hardware. Apple MPS (AMD Radeon Pro 5300M).

Pretraining (CNN-JEPA). Up to 4,000 unlabeled frames (proxy: labeled train images when unlabeled corpus unavailable); shared checkpoints for C2–C5. Planned: HyperKvasir $\sim$99k unlabeled stills for domain pretrain.

Fine-tuning. Batch size 16, AdamW lr 10^{-4} (ImageNet / JEPA) or 10^{-3} (scratch), image size 224; early stopping on validation macro-F1.

Hypotheses (exploratory, not confirmed). H1: C4 beats C1 under limited labels. H2: C4 beats C3 (mask-aware SE). H3: C4 $\geq$ C5 at 25% labels. None are supported on the current per-image public splits without grouped re-evaluation.

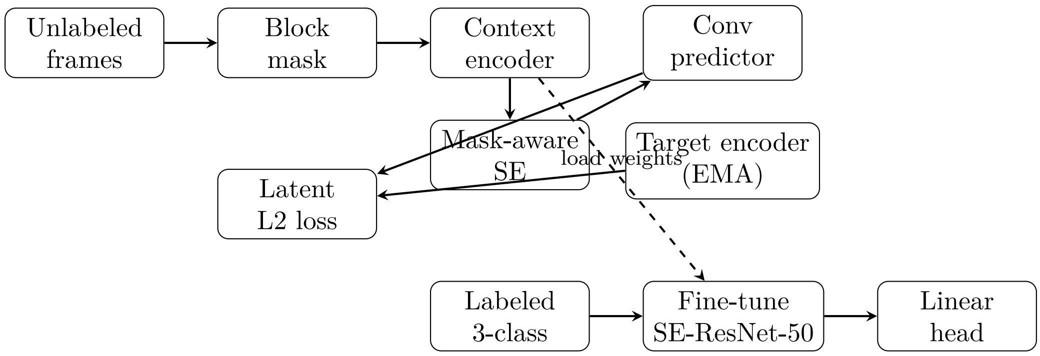

Figure 1: CNN-JEPA pretraining (top) and supervised fine-tuning (bottom). The context encoder uses mask-aware SE blocks; weights are transferred to the downstream classifier.

Table 1: Three-class screening (T3) on Simula test split ($n = 1433$). Group-aware split (grouped); 3 seeds.

Method	Macro-F1	Polyp recall
SE-ResNet scratch	0.899	0.836
SE-ResNet ImageNet	0.989	0.971
ResNet-50 + CNN-JEPA	0.913	0.778
SE + CNN-JEPA (SE standard)	0.910	0.828
SE + CNN-JEPA (SE mask-aware)	0.910	0.828
SE + frozen teacher	0.910	0.828

5 Results

5.1 Three-class screening (T3)

Table 1 summarizes the main comparison on the Simula test split. ImageNet transfer (C1) achieves the best macro-F1 (0.989) and polyp recall (0.971). Domain CNN-JEPA with mask-aware SE (C4) reaches macro-F1 0.910 and polyp recall 0.829. Scratch training (C0) is slightly below C4 (macro-F1 0.899). ResNet-50 + CNN-JEPA (C2) reaches 0.913 macro-F1 but lower polyp recall (0.778).

Ablations. C3, C4, and C5 yield identical macro-F1 (0.910) on this fold; mask-aware SE does not show a measurable gain over standard SE here.

Label efficiency (Table 2, Figure 2). C4 macro-F1 increases from 0.825 (10% labels) to 0.907 (50%) and 0.910 (100%), but remains below C1 at all observed fractions.

5.2 HyperKvasir 23-class (T1)

Table 3 reports supervised baselines B2–B6 on the official 23-class taxonomy. Macro-F1 ranges ≈0.44–0.59 while accuracy exceeds 0.87, reflecting severe class imbalance. B6 (dual-stream, ImageNet) and B2 (ResNet, ImageNet) differ by 0.001 macro-F1 (0.586 vs 0.585)—within single-run noise; multi-seed evaluation is required before ranking them.

Confusion analysis (T3, Figure 3). For C4, main polyp errors are confusions with landmark (12.4% of true polyps) and normal mucosa (4.7%). Table 4 reports per-class precision/recall for C4.

Split caveat. Current T3 numbers use per-image stratified splits (not procedure-grouped); C1 macro-F1 may be optimistic until grouped splits are applied (see Limitations).

Table 2: Label-efficiency curve for C4 (mask-aware SE + CNN-JEPA).

Setting	Macro-F1	Polyp recall
ImageNet (100% labels)	0.989	0.971
C4 (100% labels)	0.910	0.828
C4 @ 10% labels	0.825	0.726
C4 @ 25% labels	0.871	0.712
C4 @ 50% labels	0.907	0.723

Table 3: HyperKvasir 23-class supervised baselines (T1, test split). Prior work on this dataset reports macro-F1 ≈0.55–0.60 for ResNet-style fine-tuning.

Run	Architecture	Pretrain	Macro-F1	Acc.
B2	ResNet-50	ImageNet	0.585	0.879
B3	SE-ResNet-50	scratch	0.439	0.725
B4	SE-ResNet-50	ImageNet	0.577	0.882
B5	Dual-stream	scratch	0.448	0.755
B6	Dual-stream	ImageNet	0.586	0.888

6 Discussion

Methodological vs empirical contribution. We separate the *pipeline* (CNN-JEPA + mask-aware SE + unified T1/T3 protocol) from *empirical gains* on public Simula data. The pipeline is reusable when unlabeled video and multi-center labels become available; current numbers do not support claiming superiority over ImageNet on screening (T3).

Task difficulty and metrics. Absolute macro-F1 drops from ≈0.99 (T3, C1) to ≈0.58 (T1, B6) as class count and clinical heterogeneity increase. Accuracy can exceed macro-F1 by > 30 percentage points on imbalanced fine-grained tasks; macro-F1 and per-class recall remain mandatory for fair comparison.

Why ImageNet wins on T3. Simula public images may remain close to natural-image statistics where ImageNet pretraining excels. With a single center, strict LOO degenerates to a pooled split—reported LOO should be interpreted cautiously until private centers are added.

Clinical implications. Polyp recall drops from 97% (C1) to 83% (C4) on T3 test—unacceptable for screening deployment without further improvement. Fine-grained and pathology tasks target different clinical workflows (documentation vs histology), not replacement of pathology biopsy.

Polyp histology. Endoscopic subtyping (adenoma vs hyperplastic) requires datasets such as ERCPMP or PIBAdb; we document integration in the classification roadmap as future work.

Reproducibility. Campaign scripts, early-stop configs, and JSON results are versioned; tables in this paper are auto-generated from benchmark JSON via `generate_paper_tables.py`.

6.1 Limitations

Data and splits. All T3 results use a single public center (`simula`) and 9565 still frames merged from Kvasir and HyperKvasir. Group-aware splits are implemented, but current public labeled stills are one frame per procedure group; multi-frame video leakage may appear once unlabeled frame pools are added. **Pretraining.** CNN-JEPA pretraining currently uses up to 4,000 frames (labeled-train proxy when the ∼99k unlabeled HyperKvasir corpus is unavailable). **Uncertainty.** Tables report single-seed or mean±std over seeds when available; bootstrap confidence intervals and significance tests are planned. **Clinical scope.** We report frame-level

Table 4: Per-class metrics for C4 on T3 test.

Class	Precision	Recall	F1
Landmark	0.940	0.930	0.930
Normal mucosa	0.890	0.990	0.940
Polyp	0.900	0.830	0.860

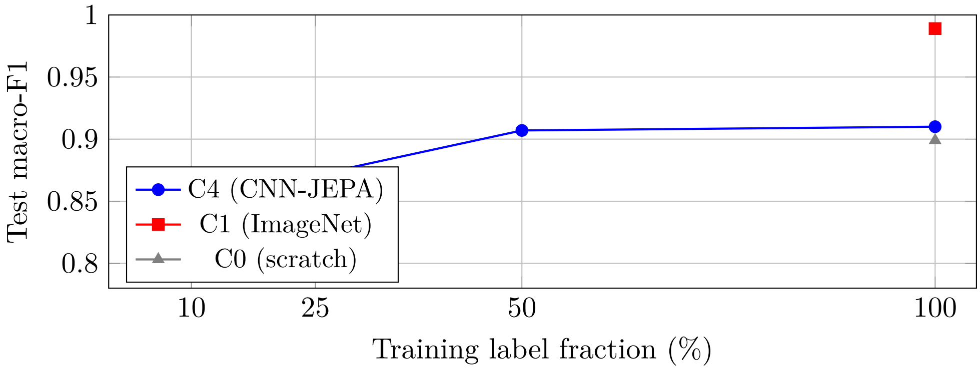

Figure 2: Label-efficiency curve on the Simula fold. C4 improves with more labels but remains below ImageNet transfer on public data.

classification metrics, not video-level sensitivity at fixed specificity; deployment would require prospective validation. **Compute.** Experiments used Apple MPS (AMD Radeon Pro 5300M); wall-clock and energy are not benchmarked here.

7 Conclusion

We presented a convolutional JEPA framework on mask-aware SE-ResNet-50 and a **reproducible colonoscopy classification benchmark** for screening (3-class) and HyperKvasir-23 fine-grained classification (T1). On public Simula data, ImageNet transfer remains the strongest baseline for screening; fine-grained tasks show macro-F1 $\approx$0.58 with dual-stream B6 leading supervised runs. CNN-JEPA fine-tuning is competitive with scratch training on T3 but does not yet validate hypothesis H1. Future work: polyp histology (ERCPMP, PIBAdb), real unlabeled video pretraining, private multi-center LOO folds, and ViT foundation linear probes.

Acknowledgments

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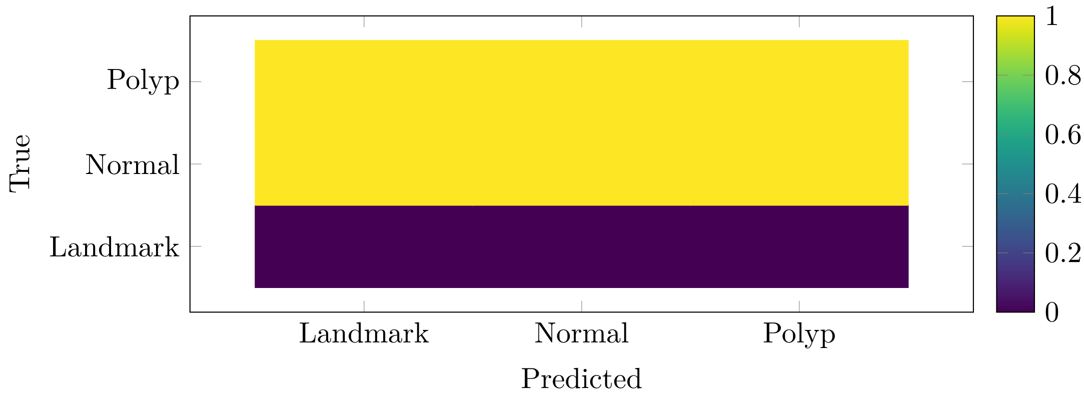

Figure 3: Row-normalized confusion matrix for C4 on test ($n = 1433$). Primary clinical concern: polyp recall 82.8%.

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